

AI & Quantum Transformation

with Prof. Anne-Marie Kermarrec, EPFL & Claudia Nemat, ABB

Board Agora Discussion Session / 21st September 2026 / Zurich

EXECUTIVE SUMMARY

Two Technologies, One Governance Question

On September 21, 2026, Board Agora convened a hybrid discussion session in Zurich to examine a shift most boards have not yet organised themselves to handle: the convergence of artificial intelligence and quantum computing, and what it means for the way directors govern technology risk and opportunity.

The session brought together board members from across European industry for a conversation co-moderated by Anne-Marie, from the research side, and Claudia, drawing on direct board and executive experience across telecommunications, industrials and financial market infrastructure.

The premise of the session was deliberately uncomfortable. Most boards have AI somewhere on the agenda. Far fewer have a clear, evidence-based picture of where the business actually stands in maturity, in governance, in dependency and fewer still have begun to ask what changes when quantum computing moves from research curiosity to practical capability. The session was framed not as a technology briefing, but as a governance one: what do directors actually need to ask, own and decide, and how does that change as the two technologies converge.

This knowledge paper distils the frameworks, tensions and arguments that emerged from that discussion.

THE STARTING POINT

Moving at Light Speed, Standing Still

The session opened with an image rather than a statistic. Modern AI and data, one moderator suggested, behave a little like the Earth and the Sun at once. The Earth moves around the Sun at thirty kilometres per second, and yet, sitting still, we notice nothing. Look up at the evening sky and the sun appears to move slowly around us. Two speeds exist in the same system, depending on where you stand.

That doubleness, it was argued, describes the current state of AI in most companies precisely. The speed is breathtaking: a new version of a leading language model arrives most weeks: the reasoning behind a model's own conclusions is becoming harder to trace as models begin optimising themselves, a phenomenon researchers are calling opaque recurrence, and warnings about the timeline to general artificial intelligence – including probability estimates of catastrophic outcomes – now arrive on a near weekly basis. Access to frontier models has itself become a geopolitical variable, with disruptions this year affecting which markets and institutions could rely on which providers.

And yet, underneath that motion, there is a standstill. Models continue to hallucinate – better than before, but not reliably enough. Companies remain unclear about what data they should actually feed these systems. And four years after generative AI reached the mass market, the practical reality for most companies is that bottom-line productivity gains and top-line growth have not materialised at the scale promised – while token costs continue to climb. More than 80% of companies, it was noted, are still not seeing the productivity gain they expected, largely because they have not asked the more precise question underneath the investment: which specific problem is this meant to solve.

"We are moving ahead at light speed – and, at the same time, standing still. That's the paradox boards need to sit with, rather than resolve too quickly in either direction."

THE RESEARCH PERSPECTIVE

Reliability, Sovereignty and the Discipline of Staying Informed

Opening from the research side, the session's academic moderator, Anne-Marie, set out three risks boards consistently underweight, before turning to what she called the discipline of staying informed.

Reliability Has Not Been Solved

Large language models still hallucinate. Alignment methods have improved the picture but have not eliminated it, and the incidents keep accumulating – including, as a recent example from legal practice illustrated, a fully invented case citation relied upon in a real legal proceeding. The message to boards was direct: this cannot simply be delegated to an agent and left unattended. Accuracy and reliability need active, ongoing verification, not a one-time sign-off.

Privacy, Sovereignty and Security Sit Together

A recent and widely discussed incident – a mathematical proof achieved with the covert assistance of AI systems trained on inputs from a private, restricted research meeting, without acknowledgement to the researchers whose work was drawn on – was offered as a caution: privacy exposure does not require an LLM specifically, and it does not always announce itself. Locally fine-tuned models, run on infrastructure a company controls, were named as one practical mitigation, alongside a broader discipline of not defaulting to full dependency on any single external provider – a question of availability and sovereignty as much as privacy. Security was added as a third, related concern: there are already documented cases of AI agents operating outside their intended boundaries, and boards need to prepare for that category of incident specifically, not only for data leakage.

The Pace Itself Is the Governance Challenge

The landscape has moved from large language models becoming mainstream only a few years ago, to agentic systems becoming mainstream more recently – and continues to shift within the lifespan of a single research grant, let alone a single strategic plan. The recommendation was not to chase every development, since not everything that generates buzz is relevant to a given business. It was to build a standing discipline of staying current: regular, structured exposure for board members – sometimes no more than a day every six months – combined with either an internal research capability or a genuine partnership with a university, so that the board has a channel to people whose job is to actually track the frontier.

"Nobody has a choice right now. This has to be embraced at every level – not as a matter of whether, but of how well you do it."

THE BOARD PERSPECTIVE

Opportunities, Risks, and Five Questions Boards Must Own

The session's practitioner moderator, Claudia, opened with a personal note: a childhood inspiration to study physics, a career that has come back to quantum through chairing the Quantum Advisory Council at a quantum computing firm, running an AI security company, and sitting on multiple boards. From that vantage point, she set out a structured view of where AI already changes the business today, what quantum specifically adds, and what she considers the non-negotiable list of questions for boards.

Where AI Already Pays Off

- Defence: AI-enabled attacks are already a daily reality – one major telecommunications network described facing roughly 50,000 attack attempts per minute, a volume that makes real-time human response impossible without AI-based pattern recognition.
- Speed: in large-scale field and Technical Service operations, language models have cut average handling time dramatically – one telecom example saw per-connection time fall from 225 seconds to 9 – addressing a genuine workforce shortage rather than only cutting cost, which changed how the change was received internally.
- Industrial competitiveness: AI is described as foundational to the autonomous systems on which Europe's industrial base increasingly depends, with real competitive pressure from faster-moving markets abroad.

Where the Risk Concentrates

- Cyber: threat sophistication is scaling with the technology itself; the coordinated, self-directed hacking behaviour recently observed among AI agents in an unrelated incident was cited as a warning of what happens when systems pursue an objective exactly as instructed, without the judgment a human would apply.
- Vendor dependency: a question that rarely reached board level in the past now needs to. Being critically dependent on a single vendor for a security- or operations-critical function is a board-level risk, not an IT decision.
- Business model disruption: boards need to ask, concretely, how exposed their own model is – while recognising that moving too fast is not automatically smarter than moving at the right pace, and that being late is equally dangerous.

The Quantum Layer

Most companies today treat quantum computing as a curiosity confined to the R&D department – and that, the session argued, is itself the mistake. By 2030, on the view of leading researchers in the field, a meaningful set of business problems will be solved not by quantum computing alone, but by hybrid workflows combining quantum with traditional high-performance computing. Two sectors stand out specifically: pharmaceuticals and materials science, where drug discovery is fundamentally a molecular simulation problem that classical simulation is reaching the limits of, and where quantum-hybrid workflows do not just accelerate discovery but improve its fidelity – with direct implications for personalised medicine; and financial services, where problems currently addressed through Monte Carlo

simulation – risk modelling, transaction pricing – hit their own computational ceiling and stand to benefit directly from hybrid approaches.

The most immediate quantum-related exposure, however, is not computational – it is cryptographic. Data encrypted securely today will not necessarily remain secure once quantum computing matures. Migrating to post-quantum cryptography was named as likely the single most fundamental infrastructure transformation ahead for the financial services sector specifically – and, implicitly, for any organisation holding data with a long confidentiality life.

“Getting ready for post-quantum cryptography is probably going to be the most fundamental infrastructure transformation the financial services sector has ahead of it.”

Five Areas Boards Need to Own

The session set out a structured framework – not a checklist to delegate, but a standing agenda for the board itself.

- **Strategy:** oversight of AI strategy and long-term value creation, with a direct question on business impact for people, processes, customers and products – and a caution that individual productivity gains do not automatically translate into company-level gains unless workflows are optimised end-to-end. Underneath all of it: is the data foundation, governance and accountability actually in place, and is investment under control.
- **Technology:** understanding – at board level, not just executive level – the company's own AI capabilities versus its outsourcing to external vendors, its cyber-defence architecture, and its critical vendor dependencies.
- **Human resources:** safeguarding the quality of decision-making and critical thinking as AI is adopted, ensuring lifelong learning and upskilling, and securing access to the right talent pipeline globally.
- **Risk and compliance:** navigating contradictory jurisdictions, staying compliant with the company's own stated values – including, explicitly, avoiding discriminatory algorithms – and safeguarding privacy to the standard the company has actually committed to, not merely the legal minimum.
- **The governance of the board itself:** whether a technology committee exists and functions without becoming a place the rest of the board delegates its thinking to, how AI is folded into risk management and audit reporting, and how the board stays current with a fast-moving regulatory backdrop – regulation such as the EU's Digital Operational Resilience Act was cited as a direct, binding example already in force.

A concrete example of putting this into practice: one organisation described building an internal control-and-routing platform connecting multiple internal and external

language models to the company's own data in an enterprise environment, so that sensitive data never has to leave the organisation's control – paired with active encouragement for board members themselves to use and test the tools employed inside the business, not merely review reports about them.

"Your accountability as board members increases. This is not something to delegate to the executive team, not just to the CTO, and not just to the one board member who happens to know more about technology than the others."

THE DISCUSSION – WHAT THE ROOM TOLD US

How Does a Board Actually Get a Feel for Where It Stands?

The floor discussion opened with a candid admission: most boards, asked directly, will say they use AI and are broadly satisfied – without having a genuine, granular picture of where and how deeply it is actually used. Two complementary answers emerged. One moderator argued for a practical, bottom-up approach: regular board-level AI training, direct board access to the same tools used inside the company, and – critically – asking pointed rather than superficial questions: adoption rates by business area, whether AI is primarily used for individual productivity or for redesigning core workflows, and how human oversight of automated decisions, such as in pricing engines, is actually being preserved as automation increases. The other moderator added a more strategic, top-down layer: understanding the split between internal-process impact and external, customer- and product-facing impact, since boards often have visibility into one and not the other.

Scientists on the Board – Asset or Distraction?

A sharp disagreement surfaced over board composition. One participant, describing direct experience recruiting deep technical experts – including in a prior wave, around cybersecurity – onto listed company boards, called the outcome an outright disaster: highly specialised experts tended to speak only within their narrow domain and struggled to contribute at the broader, business-mindset level a board actually needs. The counter-argument, however, was equally direct: the room concluded that the failure described was less an indictment of scientific expertise itself than of selecting the wrong profile, or of using permanent board seats for a need that is often episodic. The resolved view that emerged favoured a middle path: build a genuinely scientific mindset and technological fluency into the board's business-plus-technology composition but bring in deep domain specialists as invited guests for a defined problem or period, rather than installing them permanently – because boards ultimately need to operate at eye level across every topic, not defer to a single expert voice.

"You need to find the right balance between business experience and technological know-how – otherwise there is a real danger this gets treated as a silo."

A Structural Worry: Boards Narrowing to CEO and CFO Profiles

Several participants flagged a broader structural trend with direct relevance to this topic: headhunters and chairs, across many markets, are increasingly narrowing board composition toward CEO and CFO profiles specifically – a trend one moderator called genuinely dangerous, in part because that generation often built its expertise in an earlier technological era. The counter-risk was also acknowledged: narrow, purely academic technical expertise does not solve the problem either. The recommendation that gathered the most support was a shift in emphasis from CFO–CEO pairings toward CTO–CEO pairings, or deliberately building sufficient business-technology leadership into the board directly, precisely because that combination brings both the scientific literacy and the operational understanding of how technology actually gets embedded into the business.

The Case-Driven Argument: A 50-Million-Euro Refactor, and Three and a Half Hours

One of the session's most concrete illustrations came from a board member sitting on several technical-company boards. A hundred-year-old industrial testing-machine manufacturer had accumulated thirty-year-old machine-control code that no one still understood. A ten-year modernisation project to refactor it was declared failed after fifty million euros. Two years ago, an AI-based attempt also stalled. Roughly a year ago, using a more recent generation of coding model, the same code was successfully rewritten in three and a half hours – creating what the participant described as an enormous shift in momentum on that board, and a durable lesson: look for the real, unsolved, down-to-earth technical problems already sitting on the board's own agenda, and use AI as a tool against a genuine case, rather than adopting it in the abstract.

A second example, drawn from a visit to a device manufacturer's factory in China, illustrated the more disruptive end of the same logic: a company that reasoned from a blank page – how do we build a car with the software experience of one premium competitor, the driving feel of another, at a third of the cost – and then worked backward to the technology required to get there. The suggestion to boards: alongside solving known problems, periodically flip the question and ask who could attack your business at a fraction of your cost and several times your speed, and what that would require of you in response.

Vendor Dependency and the Control-and-Orchestration Layer

A recurring and pointed exchange concerned technology dependency. One moderator was direct: true independence from major technology vendors does not exist for any large company today – the realistic ambition is not zero dependency, but a deliberately owned control-and-orchestration layer sitting above the core transaction systems, connecting to multiple large language models rather than a single provider, incorporating a company's own data lineage rather than outsourcing that entirely into a vendor's black box, and maintaining at least two cloud providers rather than one. Boards, it was argued, should insist specifically that the company retains its own view of data lineage and semantics, whichever vendors it works with – not as a rejection of any specific vendor, but as a baseline discipline. The build-buy-partner framing was reframed as increasingly fluid: today's partner can become tomorrow's competitor, and the ecosystem needs to be assessed continuously from both a competitive and a co-creation lens, not settled once.

Bringing Quantum Into a Board's Regular Work – Especially in Pharma and Biotech

A participant sitting on the board of a company operating in pharmaceuticals and biologics asked the most direct operational question of the session: given that experts cannot simply be delegated to and consulted occasionally, what single practical step would actually bring AI and quantum into the board's regular working rhythm, particularly in a sector where quantum stands to make a material difference. The response converged on two points: demystifying language matters – a quick poll on what board members actually understand by terms like post-quantum cryptography would likely reveal significant gaps that need closing directly, not through embarrassment-avoidance – and internal capability matters more than periodic external expert visits: someone inside the organisation needs to be living in that technical world continuously, so the board is told when readiness is genuinely required rather than discovering it after the fact.

Measuring What AI Actually Delivers

The discussion returned repeatedly to measurement, and to a useful historical parallel: it took roughly fifty years for advertising effectiveness measurement to mature into something companies broadly accepted, even as the underlying medium kept getting more complex. AI is being asked to prove its return within a single budget cycle – a standard, several participants acknowledged including a self-critical admission, that companies themselves have pushed for. The practical distinction that emerged was to measure differently by domain rather than seeking one universal KPI: AI for people should be measured through engagement, learning and adoption metrics rather than direct productivity, because building genuine workforce capability takes time and tolerance; AI for core business processes, particularly

production, is far more directly measurable, and failure there is immediately visible in a way that failure in, for instance, HR often is not; and AI viewed through the lens of the customer is a distinct, strategic question again, requiring its own metrics rather than being folded into the other two.

"It's an outcome discussion, not an input discussion. It's very easy to say here is my AI budget and here is what I spent. The harder – and more important – question is what you actually got for it."

GOVERNANCE ARCHITECTURE

A Standing Capability, Not a Standing Start

A thread ran quietly under almost every part of the session, and it is worth pulling out on its own: this is not a technology that arrives, gets adopted, and is then done. It is a continuous transformation – the pace itself was the point made repeatedly – and boards that treat each new wave as a fresh, standalone adoption project will keep paying the cost of starting from zero, every time.

The alternative that emerged from the discussion was not a single fix but a set of standing mechanisms, each one designed to carry forward rather than reset.

- A cadence, not a briefing. Board-level exposure to what is actually changing needs to be scheduled and recurring – a few days every six months was the working suggestion – rather than a one-off training delivered when a new technology first becomes visible. The point is not to catch up once; it is to never fall far behind again.
- A method that gets reused, not reinvented. The case-driven approach described in the room – start from a genuine unsolved business problem, apply the right tool, learn from the result – is only valuable if it becomes a standing discipline the board returns to for the next wave, rather than a one-time success story. The same applies to the shift from measuring input to measuring outcome: once a board has built that discipline for AI, it should not need to rebuild it from scratch for quantum.
- An architecture built to absorb change, not to be replaced by it. The control-and-orchestration layer described earlier – sitting above core transaction systems, connecting to multiple models and providers rather than one – was framed specifically as infrastructure that lets a company adopt the next capability without renegotiating its entire technology stack each time. The alternative, seen repeatedly in the room, is companies whose past investments become the very thing constraining their next move.

- A standing relationship with expertise, not an occasional consultation. Whether through an internal research function or a genuine university partnership, the value is in continuity: someone inside or close to the organisation who is already living in the next wave before the board needs to ask about it, rather than a new search for expertise each time a new term enters the boardroom.
- A board composition designed for continuity of judgment. The recurring preference for business-technology fluency built into the board itself – rather than relying solely on episodic outside experts – reflects the same logic: capability that stays in the room compounds; capability that is invited in and out does not.

None of this removes the need to evaluate each new technology on its own terms – AI and quantum are genuinely different, as the session was careful to establish. But the muscle for evaluating, piloting and absorbing a new technological capability is the same muscle each time. Boards that build it once, deliberately, spend the next wave adapting. Boards that do not, spend it catching up – and quantum will not be the last wave to test which kind of board they are.

THE CLOSING ARGUMENT

Curiosity, Not Delegation

Asked to distil two things to take back to her own board, Anne-Marie returned to the theme that had opened the discussion: this is a strategic matter, not a technology issue, and boards should resist any temptation to delegate it – to the executive team generally, to the CTO specifically, or to a single technically literate board member. Accountability, she argued, increases for every director as these technologies become more consequential, not less; and that accountability requires genuine judgment and a willingness to keep learning, not a one-time briefing.

Her closing note turned toward tone rather than mechanics: despite the genuine difficulties, this is – for boards and for anyone working in the field – an unusually exciting moment to be doing the work. The invitation to directors was to bring a spirit of curiosity to their engagement with the executive team on these questions, rather than treating the topic purely as a risk to be contained – and, in doing so, to help move the conversation out of the risk corner and further into the innovation corner that Europe, in particular, needs more of.

“Adopt a bit of that spirit of curiosity and excitement and bring it into your interactions with the executive team – that’s what drives things out of the risk corner and into the innovative one.”

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This knowledge paper synthesises insights from the Board Agora discussion session held on 21 September 2026. We extend our gratitude to Prof. Anne-Marie Kermarrec and Claudia Nemat for their expert contributions, and to all participating board members whose candid reflections and challenges enriched this dialogue. The quality of a Chatham House conversation depends entirely on the willingness of participants to speak with genuine honesty.

Board Agora Knowledge Paper

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